

Expression analysis of calcium-dependent protein kinase gene family during reproductive development and abiotic stress conditions in rice (*Oryza sativa* L. ssp. *indica*)

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Abstract Calcium-dependent protein kinases (CDPKs) are important sensors of Ca^{+2} flux in plants, which control plant development and responses by regulating downstream components of calcium signaling pathways. Availability of the whole genome sequence and microarray platform allows investigation of genome-wide organization and expression profile of CDPK genes in rice with a view to ultimately define their function in plant systems. Genome-wide analysis led to identification of 31 CDPK genes in rice after a thorough annotation exercise based upon HMM profiles. Twenty-nine already identified CDPK genes were verified and two new members were added to the CDPK gene family of rice. Relative expression of all these genes has been analyzed by using Affymetrix rice genome array™ during three vegetative stages, six stages of panicle (P1–P6) and five stages of seed (S1–S5) development along with three abiotic stress conditions, viz. cold, salt and desiccation, given to seedling. Thirty-one CDPK genes were found to express in at least one of the experimental stages studied. Of these, transcripts for twenty three genes accumulated differentially during reproductive developmental stages; nine of them were preferentially up-regulated only in panicle, five were up-regulated in stages of panicles as well as seed development, whereas, expression of one gene was

found to be specific to the S1 stage of seed development. Eight genes were found to be down-regulated during the panicle and seed developmental stages. Six CDPK genes were found to be induced while the expression of one gene was down-regulated under stress conditions. The differential expression of CDPK genes during reproductive development and stress is suggestive of their involvement in the underlying signal transduction pathways. Furthermore, up-regulation of common genes both during reproductive development as well as stress responses is indicative of common element between reproduction and stress.

Keywords Calcium-dependent protein kinase · *Oryza sativa* ssp. *indica* var. IR64 · Microarray · Expression profile

Introduction

In plants, transient changes in calcium concentration in the cytoplasm in response to extrinsic and intrinsic cues are sensed and decoded by an array of calcium sensors, viz., calmodulins, calmodulin-like proteins, calcineurin B-like proteins and calcium-dependent protein kinases (Snedden and Fromm 1998, 2001; Trewavas and Malho 1998; Reddy 2001; Luan et al. 2002; Sanders et al. 2002). The calcium-dependent protein kinases (CDPKs) constitute one of the largest calcium sensing subfamilies of plant-specific protein kinases that are reported to be found throughout the plant kingdom from algae to angiosperms (Ludwig et al. 2004) along with a few reports that confirm their presence even in protozoa (Harmon et al. 2001; Harper and Harmon 2005; Ishino et al. 2006). The CDPKs contain four characteristic domains; an N-terminal domain, a kinase domain, an autoinhibitory domain and a calmodulin-like domain

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(Harper et al. 1991; Harmon et al. 2000; Hrabak et al. 2003). The N-terminal domain is highly variable and contains myristoylation or palmitoylation sites for subcellular targeting (Cheng et al. 2002). The kinase domain is the catalytic domain with an ATP binding site, which is followed by an autoinhibitory domain (Harmon et al. 1994) and the calmodulin-like domain that contains EF-hands for binding to calcium (Cheng et al. 2002). CDPKs are known to be directly activated by binding to calcium (Sanders et al. 2002) via calmodulin-like domain leading to a conformational change in the kinase domain, which results in the displacement of the autoinhibitory domain, known otherwise to engage the active site as pseudosubstrate (Reddy 2001).

After the availability of complete *Arabidopsis* genome sequence, a comprehensive exercise to collate the information on all 34 CDPK encoding genes was made (Cheng et al. 2002; Hrabak et al. 2003). Similar attempts in rice revealed 29 CDPK genes by using an earlier version of the genome sequence (Asano et al. 2005). A comparative phylogenetic analysis of rice and *Arabidopsis* CDPK superfamily revealed seven groups, where CDPKs were distributed in four subgroups and CDPK-related kinases (CRKs), calcium- and calmodulin-dependent protein kinases (CCaMKs) and phosphoenolpyruvate related kinases (PEPRKs) were clustered in three distinct classes. Since rice and *Arabidopsis* CDPKs are represented in all four subgroups, there may not have been any major evolutionary change in CDPK population after the divergence of monocots and dicots, indicating their involvement in essential cell functions.

CDPKs have been reported to be involved in diverse functions in plant signal transduction (Ludwig et al. 2004). Their signal cascades influence pollen tube elongation (Estruch et al. 1994), development of root cells and root hair (Ivashuta et al. 2005), cell division, differentiation and programmed cell death (Yoon et al. 1999; Lee et al. 2003), stress response (Sheen 1996) and control of NADPH activity (Xing et al. 2001). In rice, *OsCPK13* (*OsCDPK7*) and *OsCPK7* (*OsCDPK13*) have been implicated in stress signaling (Saijo et al. 2000; Yang et al. 2003; Abbasi et al. 2004) and *OsCPK23* (*SPK*) and *OsCPK19* (*OsCDPK2*), have been reported to be essential for seed development (Kawasaki et al. 1993; Breviario et al. 1995; Frattini et al. 1999; Morello et al. 2000). The *OsCPK23* has further been shown to act as sucrose synthase kinase that regulates supply of sucrose for biosynthesis of storage products (Asano et al. 2002). The expression patterns of certain CDPK genes have been investigated in maize, rice, alfalfa, *Arabidopsis*, cucumber, ice plant, mung bean, potato, strawberry, tobacco, tomato, *Medicago* and found to be regulated by fungal elicitors, abiotic stresses, hormones and the intrinsic developmental program (Urao et al. 1994; Monroy and Dhindsa 1995; Botella et al. 1996; Berberich and Kusano 1997; Yoon et al. 1999; Patharkar and Cushman 2000;

Davletova et al. 2001; Murillo et al. 2001; Raices et al. 2001; Chico et al. 2002; Llop-Tous et al. 2002; Ullanat and Jayabaskaran 2002; Akimoto-Tomiya et al. 2003; Raices et al. 2003; Chehab et al. 2004; Harper et al. 2004; Kumar et al. 2004; Asano et al. 2005; Ivashuta et al. 2005; Szczegieliński et al. 2005; Zhang et al. 2005; Furutani et al. 2006). It may, however, be noted that genome-wide expression analysis of CDPK gene family for any species is still lacking. The availability of complete genome sequence of rice (International Rice Genome Sequencing Project 2005) and the high-throughput expression analysis method like microarray have enabled us to do genome-wide analysis to help undertake functional characterization of all the genes belonging to this class of kinases.

In this report, an analysis of the expression profiles of 31 CDPK genes that were identified by a comprehensive bioinformatics analysis of rice genome sequence database is presented. Expression analysis using Affymetrix 57K GeneChip®, during six stages of panicle and five stages of seed development in comparison to leaf, root and seedling has revealed a number of genes that express specifically or are common to these developmental stages. Transcripts of seven genes were found to accumulate differentially during cold, salt and desiccation stress.

Materials and methods

Identification of calcium-dependent protein kinase genes in rice

For the identification of CDPK homologues in rice, 153 CDPK protein sequences were searched from the NCBI (National Centre for Biotechnology Information; <http://www.ncbi.nlm.nih.gov/BLAST/>) database of various species. An HMM (Hidden Markov Models) profile was prepared using HMMER program (<http://hmmer.wustl.edu/>) (Eddy 1998). TIGR (The Institute for Genomic Research; <http://www.tigr.org/>) version 4 pseudomolecule database and the KOME (Knowledge-based Oryzae Molecular biological Encyclopaedia; <http://cdna01.dna.affrc.go.jp/cDNA/>) full-length cDNA database were searched for the probable CDPK protein sequences. The probable CDPK protein sequences were subjected to PROSITE scan (Falquet et al. 2002) for their domain structure. The predicted CDPK gene, cDNA and protein sequences were extracted from the corresponding TIGR gene locus IDs. The locations of the cDNAs were obtained from the genome browser portal on TIGR rice database and confirmed by using BLAST (Basic Local Alignment Search Tool) version 2.0.3 (Altschul et al. 1990) against the TIGR rice pseudomolecule database. The identity and chromosomal location of all the short-listed CDPK genes was also verified in the recently released

version 5 of rice pseudomolecule database; the structural characteristics of CDPK genes were found to be the same as in version 4. The molecular masses of the CDPKs were calculated by using GENERUNNER (version 3.05). The *Arabidopsis* CDPK protein sequences were obtained from the *Arabidopsis* Information Resources (TAIR) (Garcia-Hernandez et al. 2002). The protein sequences of alfalfa-*MsCPK3* (X96723); cucumber-*CsCDPK5* (AY027885); ice plant-*McCPK1* (AF090835); mung bean-*VrCDPK1* (VRU08140); potato-*StCDPK1* (AF115406) and *StCDPK2* (AF418563); strawberry-*FaCDPK1* (AF035944); tomato-*LeCDPK1* (AF363784); *Petunia-PiCDPK1* (DQ147913) and *PiCDPK2* (DQ147912); maize-maize CDPK (L27484), *ZmCDPK1* (D84408), *ZmCPK11* (AY301062) and *ZmCPK10* (ZMA7366); tobacco-*NtCDPK1* (AF072908), *NtCDPK2* (AJ344154), *NtCDPK3* (AJ344155), *NtCPK4* (AF435451) and *Medicago-MtCDPK1* (AY821654) were downloaded from the NCBI database. For gene duplication information on the segmental duplicated regions of rice and *Arabidopsis* genome, we referred to the TIGR database.

Phylogenetic and motif analysis

The amino acid sequences of the rice CDPKs were deduced from the putative ORFs by using GENERUNNER (version 3.05). The protein sequences were aligned using ClustalX version 1.83 (Thompson et al. 1997). A protein phylogenetic tree was created according to the neighbor-joining method clustering strategy (Saitou and Nei 1987) in ClustalX program and analyzed using TREEVIEW 1.6.5 software. To find conserved motifs in CDPK group of the proteins Multiple EM for Motif Elicitation (MEME) version 3.5.4 (Bailey and Elkan 1994) was used. The parameters of this analysis were set up as: number of repetition, any; maximum number of motifs, 35; and optimum width of the motif, ≥ 6 and ≤ 300 . The motif profiles thus obtained have been schematically represented in supplementary figure S1.

Plant material and growth condition

Different developmental stages of rice panicle were obtained from field grown rice (*Oryza sativa* ssp. *indica* var. IR64). The young panicles were taken out from the sheath and measured to be categorized in six groups (P1, 0–3 cm; P2, 3–5 cm; P3, 5–10 cm; P4, 10–15 cm; P5, 15–22 cm and P6, 22–30 cm) based on length of the panicle and the landmark developmental events (Itoh et al. 2005) and frozen in liquid nitrogen. The rice seed were tagged from day of pollination (0 DAP) and seeds were collected on each DAP from 0 to 30 DAP. These were pooled into S1, S2, S3, S4, and S5, representing, 0–2 DAP, 3–4 DAP, 5–10 DAP, 11–20 DAP and 21–29 DAP, respectively.

Mature leaves were harvested from the same plants. After disinfection with 0.1% HgCl_2 and thorough washing with RO (reverse-osmosis) water, seeds were soaked overnight in RO water. Next day, seeds were spread on a meshed float and grown hydroponically at $28 \pm 1^\circ\text{C}$ in culture room with a daily photoperiodic cycle of 14 h light and 10 h dark. After 7 days of growth, the seedlings were given different stress treatments (Mukhopadhyay et al. 2004). Specifically, for cold stress the seedlings were kept at $4 \pm 1^\circ\text{C}$ in RO water, for salt stress the seedlings were transferred into a beaker containing 200 mM NaCl solution and for desiccation stress they were air-dried on a Whatman 3 mm sheet at $28 \pm 1^\circ\text{C}$ for 3 h. For control, 7-day-old seedlings were maintained in water in 100 ml beaker for 3 h. Roots were harvested from 7-day-old light grown seedlings grown in RO water.

Affymetrix GeneChip hybridization and data collection

Total RNA was isolated from vegetative tissue of 7-day-old rice seedlings, the stress samples, mature leaf and 7-day-old seedling roots and panicle of rice and quality of the RNA was ascertained as described previously (Jain et al. 2006). For isolating total RNA from rice seed, RNA isolation method from carbohydrate-rich seeds was followed (Singh et al. 2003). The microarray analysis using Affymetrix GeneChip® Rice Genome Array was carried out according to Affymetrix manual for one-cycle target labelling and control reagents (Affymetrix, Santa Clara, CA, USA) using 5 μg of RNA (Jain et al. 2007) as starting material. Target preparation, hybridization to arrays, washing, staining and scanning were carried out as described earlier (Jain et al. 2007). The cell intensity data files generated by the GeneChip Operating Software (GCOS 1.2) software (Schadt et al. 2001) were imported in Avadis 4.2 software (Strandgenomics, Bangalore, India) for subsequent data processing. The data from 51 chips were normalised by using GCRMA algorithm (Wu et al. 2004). The correlation between the biological replicates were assessed using Pearson's correlation coefficient (R) on the signal intensities and the R values between the three replicates were ≤ 0.95 for 17 experimental stages. The normalized data were subjected to $\log_2$ transformation for calculating differential expression of the experimental tissue with suitable controls. A gene was designated as up- or down-regulated if the signal ratios were ≥ 2 at P values < 0.05 with respect to mature leaf tissue sample in case of developmental stages and 7-day-old seedling for stress samples. Subsequently, a CDPK gene family specific sub-dataset was prepared and respective log transformed intensity values were used for hierarchical clustering by using Euclidean distance matrix. Hierarchical clustering was performed with log ratio transformed average values taking mature leaf and 7-day-old seedling as

baseline control for developmental stages and abiotic stresses, respectively. Further analyses were carried out in Microsoft Excel. Microarray data from this article have been deposited in the Gene Expression Omnibus database at the National Center for Biotechnology Information under the series accession numbers GSE6893 and GSE6901.

Quantitative real-time PCR

To confirm the expression patterns of CDPK genes obtained by microarray based transcriptome profiling, Q-PCR analyses were also performed on representative genes (Jain et al. 2006). The primer sequences used in this analysis are given in supplementary table S1. *Actin* was used as endogenous control to normalize variance in the quality of RNA and the amount of cDNA used. The highest expressing sample accrued the value 1 in the relative mRNA calculation using the formula $2^{-\Delta\Delta CT}$ for both the biological replicates, therefore, the bar representing highest relative mRNA value bears no error bar. These data were further extrapolated to match the amplitude of profiles with those obtained from microarray analysis. Each experiment was repeated at least two times and representative data are shown.

Results

Identification of calcium-dependent protein kinase genes in rice

In this study, genome-wide analysis of CDPK gene family has been performed on the basis of recently completed rice genome sequence (International Rice Genome Sequencing Project 2005), its updated versions and KOME full-length cDNA database. The HMM (Hidden Markov Model) profile based search for the probable CDPK protein sequences revealed 1,693 non-redundant hits. The domain analysis of the probable CDPK protein sequences using PROSITE scan (Falquet et al. 2002) revealed 31 proteins with characteristic domains. Since during the preparation of this manuscript TIGR rice pseudomolecule release five also became available, the structure and chromosomal location of CDPK gene family members was verified in the new version. Some minor changes in the position of four genes were observed and were incorporated accordingly in supplementary table S2.

Chromosomal localization of the identified genes revealed that chromosome 3 and 7 bore the maximum number of CDPK genes, i.e., five each, followed by chromosome 12 on which 4 genes were localized (Fig. 1). Chromosome 1, 2 and 5 were found to have three CDPK genes each, whereas, two CDPK genes each were found on

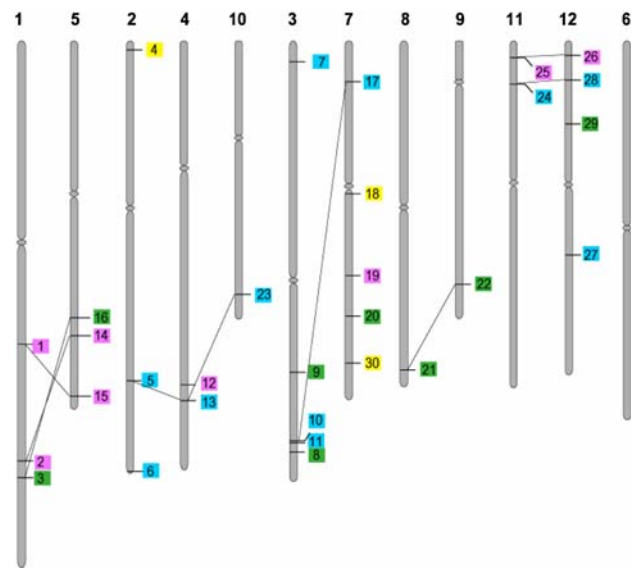
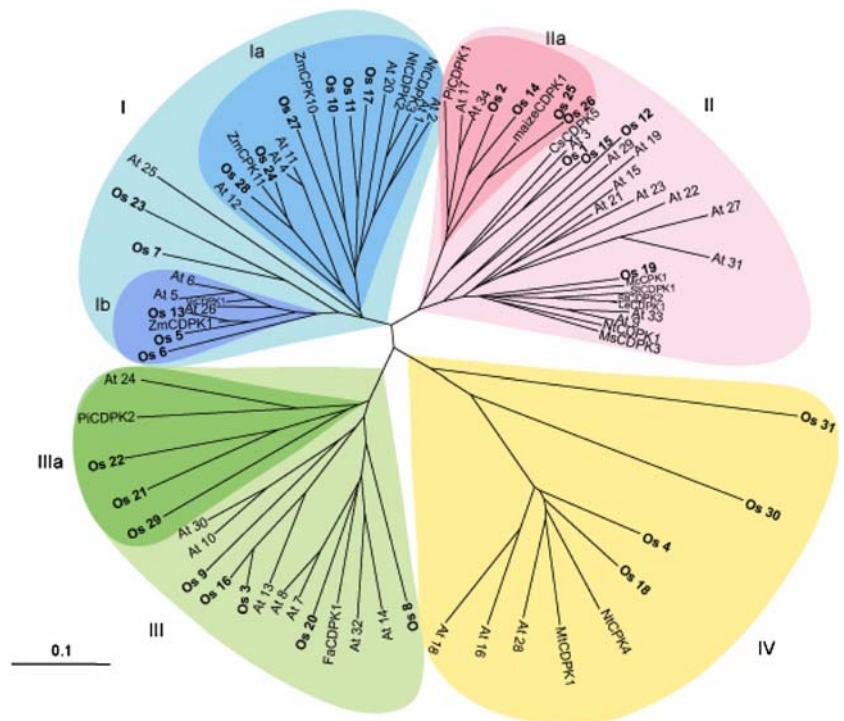


Fig. 1 Genome-wide distribution of *OsCPK* genes on the rice chromosomes. The position of centromeres are marked with grey ovals. Straight lines are drawn to connect the duplicated genes. Gene names are represented by corresponding numbers as described (Asano et al. 2005). The color boxes are indicative of the phylogenetic group shown in Fig. 2. The chromosome numbers are indicated at the top of each bar. Chromosomes have been grouped to represent duplications with clarity

chromosomes 4 and 11. Only single CDPK genes were found on chromosome 8, 9 and 10. Chromosome 6 did not have any genes of CDPK class. The *OsCPK31* could not be placed on any chromosome and it may belong to some gap regions in pseudomolecules. The superimposition of segmental duplication data from TIGR website on the dataset of chromosomal localization of CDPK genes revealed that *OsCPK1* and 15; *OsCPK2* and 14; *OsCPK3* and 16; *OsCPK5* and 13; *OsCPK13* and 23; *OsCPK11* and 17; *OsCPK21* and 22; *OsCPK25* and 26 and *OsCPK24* and 28 formed 9 gene pairs which may have resulted because of the segmental duplication of the rice genome.

To update the work of Asano et al. (2005) and to study the evolutionary relatedness of rice and *Arabidopsis* CDPKs with all the CDPK genes characterized so far from alfalfa, cucumber, ice plant, mung bean, potato, strawberry, tomato, *Petunia*, maize, tobacco and *Medicago*, an unrooted tree was constructed by using ClustalX 1.83. This exercise resulted in four distinct groups similar to that reported by Asano et al. (2005) (Fig. 2). To understand the basis of divergence between the phylogenetic groups MEME tool for motif elicitation was employed. Supplementary figure S1 shows distribution of conserved motifs in all groups of CDPK genes. Within the four groups formed, group I, II and III were found to be more closely related to each other in comparison to group IV. The group IV protein contained three motifs, #15 (75 residues within the calmodulin-like domain), #20 (42 residues in the kinase domain)

Fig. 2 Phylogenetic relatedness among the rice, *Arabidopsis* and functionally characterized CDPKs from other plant species. The phylogenetic tree was created using ClustalX 1.83, based on the predicted amino acid sequences. The four distinct phylogenetic groups are marked from I to IV. Subclades Ia, Ib, IIa and IIIa correspond to similarity in expression patterns. The branch lengths are proportional to divergence, with the scale of “0.1” representing 10% change. At *Arabidopsis* CPK and Os *Oryza sativa* CPK (e.g., At 1 AtCPK1 and Os 1 OsCPK1). The rice CDPKs are marked with **bold letters**



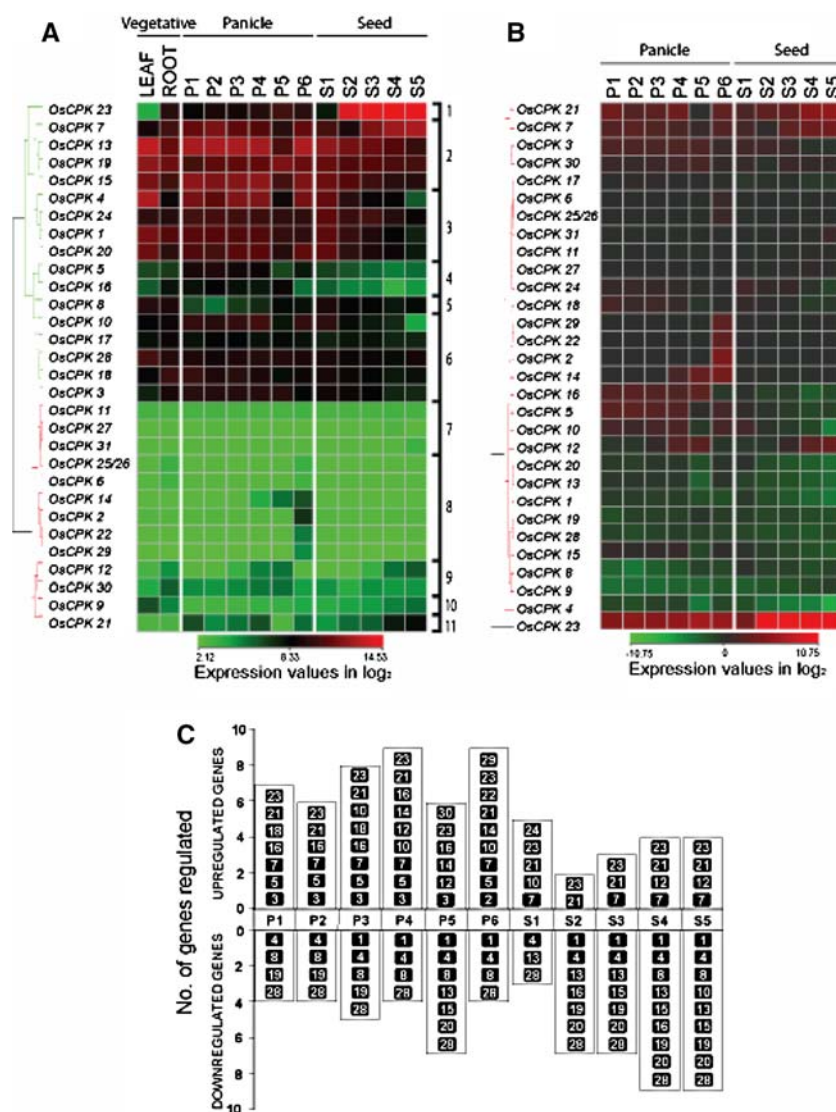
and #30 (11 amino acid residues within N-terminal domain), which were specific to this group. Some minor differences were also observed between groups I, II and III, e.g., motif #19 (21 residues within the calmodulin-like domain) was specifically found in 16 members of group III. Motif #23 (21 amino acid residues within the N-terminal domain) was predominantly found in the members of group II. Each phylogenetic group seems to have undergone expansion by segmental genome duplication events in both rice and *Arabidopsis*, which is evident from nine and eight pairs of genes localized on duplicated region of the respective genomes. In case of *Arabidopsis*, tandem duplication events resulting in five genes (*AtCPK21*, 22, 23, 27 and 31) seem to have added to the expansion of the gene family after the divergence of monocots and dicots.

Expression profiling of CDPK gene family by using microarrays

All 31 CDPK genes were found to be represented on the GeneChip. *OsCPK25* and 26, however, were represented by a common probe set because of high level of homology between the two genes; therefore, the data represents collective expression profiles of both the genes. The initial analysis revealed that all 31 genes expressed in at least one of the experimental stages analyzed in this study, which included three vegetative stages, eleven stages of panicle and seed development and three stress conditions. On the basis of hierarchical clustering, which is based on the mean log transformed intensity value of the genes, expression

profiles of these genes were grouped into eleven categories (Fig. 3a). Category 1 is represented by a single gene, *OsCPK23*, which shows up to 87- and 1,724-folds up-regulation in panicle and seed developmental stages, respectively, in comparison to mature leaf. It also shows higher expression (53-folds) in roots. Four genes *OsCPK7*, 13, 19 and 15 constitute the second category. These genes show near constitutive expression except *OsCPK7*, which is up to 6- and 14-folds up-regulated in panicles and seeds, respectively. Conversely, *OsCPK13*, 19 and 15 are down-regulated by >2-folds during late stages of seed development. *OsCPK13* and 15 are down-regulated also during P5 stage of panicle development. The third category genes, consisting of *OsCPK4*, 24, 1 and 20 show high transcript accumulation during stages of vegetative and panicle development. However, these genes are characteristically down-regulated during late stages of seed development. Two genes (*OsCPK5* and 16) belonging to category 4 are up-regulated by up to 6-folds during panicle development but are down-regulated during seed development. *OsCPK8* (category 5) transcripts accumulated at relatively higher levels in vegetative and seed developmental stages; however, in stages of panicle development, its transcript level declines up to 15-folds. Five genes belonging to category 6 show almost similar expression pattern as those of category 2, but these genes expressed at much lower amplitude. Category 7 (*OsCPK11*, 27 and 31) genes express at almost undetectable levels in all the stages analyzed, but it is possible that these genes might respond to specific stimuli or their expression might be limited to specialized cell types that

Fig. 3 Hierarchical clustering and differential expression analysis of the CDPK genes in two vegetative tissues (LEAF: mature leaf; ROOT: 7-day-seedling root), six stages of panicle development (P1: 0–3 cm; P2: 3–5 cm; P3: 5–10 cm; P4: 10–15 cm; P5: 15–22 cm and P6: 22–30 cm) and five seed developmental stages (S1: 0–2 DAP; S2: 3–4 DAP; S3: 5–10 DAP; S4: 11–20 DAP and S5: 21–29 DAP). **a** Genes with similar expression profiles are clustered in 11 groups (1–11). The color scale is given in $\log_2$ intensity values. **b** Hierarchical cluster showing average log ratio transformed intensity value of the CDPK genes in six stages of panicle and five stages of seed development, with respect to mature leaf. **c** CDPK genes showing differential up- and down-regulation during panicle and seed developmental stages. A gene is considered significantly regulated if ≥ 2 -folds change in expression has occurred at P value ≤ 0.05 with respect to mature leaf. Numbers in black boxes represent serial numbers of the genes as described in table S2



have not been analyzed in this investigation. Characteristically, all the genes (*OsCPK25/26*, 6, 14, 3, 22 and 29) comprising category 8 express during the P6 stages of panicle development. The transcripts of *OsCPK14*, however, start to show the upward trend from P5 stage. Category 9 genes, *OsCPK12* and 30, express at low levels in leaf and during last stage of panicle development. Their expression during stages of seed development is low except for *OsCPK12*, which also shows increased transcript accumulation during S4 (5-folds) and S5 (8-folds) stages of seed development. *OsCPK9* (category 10) is one of the few genes that expresses preferentially in vegetative tissues. Its highest expression was observed in mature leaves, with undetectable levels in panicle stages (supplementary table S3). However, its transcripts were also detected in late stages of seed development, the transcript levels in seeds, however, were approximately half of the values in comparison to mature leaf. The category 11 gene, *OsCPK21*, did not show

any expression in vegetative tissues, however, its transcript increased 12- and 44-folds during panicle and seed development stages, respectively.

Genes that show differential accumulation of transcripts during various stages of reproductive development in comparison to leaf have been summarized as Fig. 3b and c. For detailed fold change values and average log ratio transformed values refer to supplementary table S4 and S5, respectively. Fourteen genes were up-regulated during stages of panicle development, whereas, five were found to be down-regulated by >2 -folds. Conversely, only six CDPK genes were found to be up-regulated during seed development stages, whereas, the number of down-regulated genes was 10. Expression pattern of five genes (*OsCPK2*, 14, 21, 23 and 29) showing specific expression pattern were also verified by Q-PCR analysis. The result showed good correlation with those obtained from the microarray analysis (Fig. 4). In our analysis, category 8 genes were

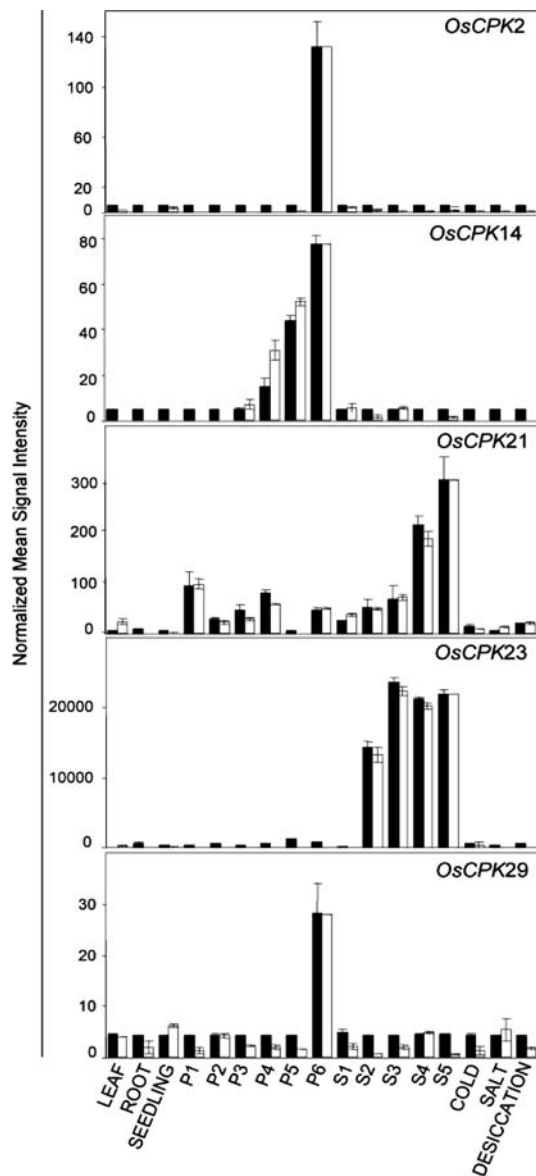


Fig. 4 The confirmation of microarray based expression profiles by Q-PCR. Expression profiles of five differentially expressed genes are shown. The filled bars are the mean normalized signal intensity value of biological replicates in microarray experiment. The hollow bars are means of relative fold change value of two biological and six technical replicates obtained by Q-PCR scaled to microarray signal intensity values. Standard error of the biological replicates are shown as error bars

found to express specifically during P6 stage of panicle development, which corresponds to mature pollen stage prior to anthesis. In the phylogenetic tree, these genes are distributed in subclades marked as IIa and IIIa.

Involvement of CDPK genes in abiotic stress

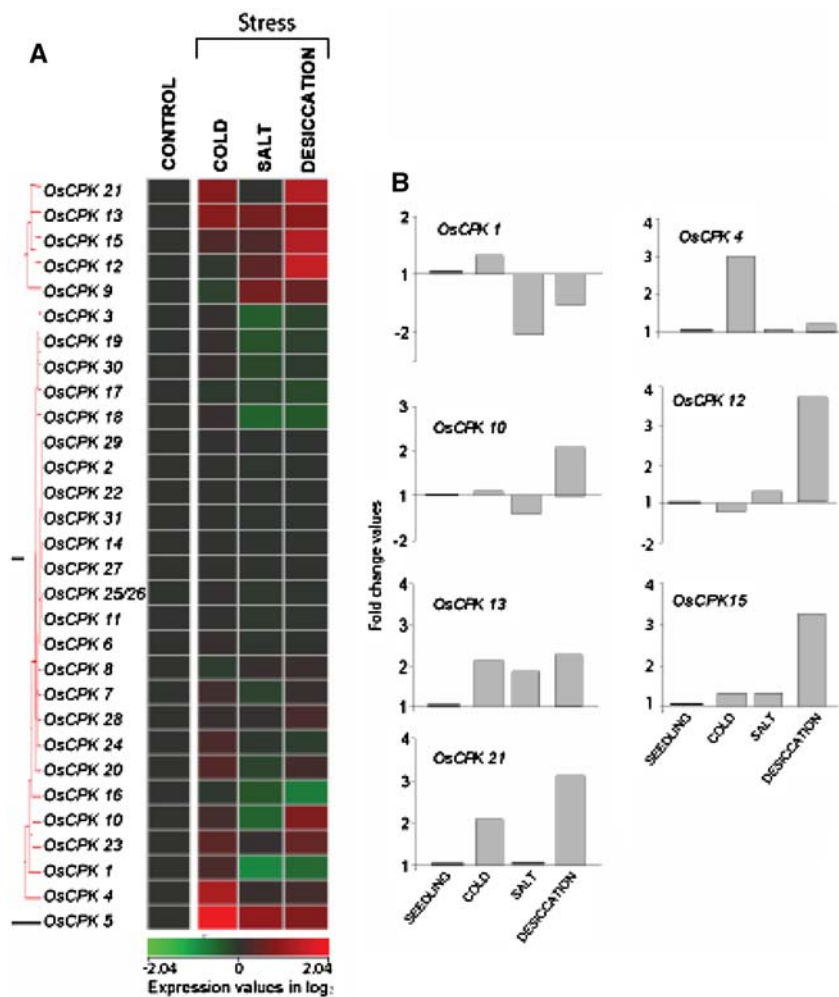
The expression of seven rice CDPK genes was found to be significantly affected in response to cold, salt and/or desiccation stress conditions (Fig. 5a). For detail average log

ratio transformed values refer to supplementary table S6. *OsCPK4* and *13* were found to be induced by 3.02- and 2.14-folds, respectively, under cold stress conditions (Fig. 5b). Five genes *OsCPK10*, *12*, *13*, *15* and *21* were found to be up-regulated under conditions of desiccation stress. The enhancement in expression of *OsCPK12*, *15* and *21* was between 3.23- and 3.77-folds, whereas, for *OsCPK10* and *13* it was 2.03- and 2.33-folds, respectively, in comparison to the non-stressed 7-day-old seedlings. None of the CDPK genes were up-regulated under conditions of salt stress at >2-folds, instead, the accumulation of *OsCPK1* transcripts declined by >2-folds when the seedlings were subjected to 200 mM salt stress for 3 h. Of the stress responsive genes, *OsCPK13* was induced under all three abiotic stress conditions. Under cold and desiccation stress its transcripts increased by 2.14- and 2.23-folds, respectively. However, under salt stress conditions the increase in transcript levels was only 1.84-folds (Supplementary table S7). To understand the regulatory mechanism of the CDPK genes which showed accumulation of transcripts under abiotic stress conditions, promoter analysis of 1 kb upstream region from the translation start site using PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) was performed. It was found that all the seven genes contained stress responsive *cis*-regulatory elements (viz. ABRE, Hattori et al. 1995; MBS, Yamaguchi-Shinozaki and Shinozaki 1993; TC-rich repeats, Diaz-De-Leon et al. 1993 and LTR, Dunn et al. 1998).

Comparison of expression profiles of duplicated CDPK genes in rice

The phylogenetic analysis of the rice, *Arabidopsis* and other characterized CDPK genes of other plant species indicated towards the expansion and diversification of the gene family by multiple level of duplication. To study the role of gene duplication in evolution and diversification of gene function, the expression profiles of the 17 CDPK genes localized on the duplicated segments of rice genome were analyzed (Fig. 6). Four gene pairs *OsCPK5:13* (84% identity), *OsCPK1:15* (86.5% identity), *OsCPK2:14* (87.4% identity) and *OsCPK3:16* (91.8% identity) were found to have retained their expression patterns after the duplication events, however, the overall amplitude of expression showed marked differences amongst the pair partners. This could indicate retention of gene function. Another set of three CDPK gene pairs, *OsCPK13:23* (61.2% identity), *OsCPK21:22* (64.6% identity) and *OsCPK24:28* (86.0% identity), showed completely divergent expression profile. The comparison of their sequences revealed much reduced level of homology at amino acid level, indicating that these genes might have undergone significant diversification after the duplication of the respective genomic segments leading to neofunctionalization for the pair partners. In the third

Fig. 5 CDPK gene expression profiles in seedlings in response to stress conditions with respect to un-stressed 7-d-old seedling. **a** Hierarchical clustering of the CDPK genes under cold, salt and desiccation. Color scale for average log ratio values is given in $\log_2$. **b** A bar graph representation of significantly differentially expressed genes under stress conditions. The bars represent transcript level fold change value with respect to un-stressed 7-day-old seedling. The positive scale represents up-regulation and negative scale represents down-regulation of transcripts under stress conditions



category, one of the duplicated gene exhibited near complete absence of expression in the stages analyzed, indicating, pseudogenization after duplication. Pseudogenization was also observed between *OsCPK17:11*, where *OsCPK11* might have lost its function during course of evolution. It was observed that *OsCPK30*, with 1 EF-hand, had a duplicated counterpart on chromosome 3, *OsCRK1* (CDPK-related protein kinase), not bearing EF-hands (Asano et al. 2005). Interestingly, it was found that *OsCPK30* expressed differentially in root tissue and late panicle developmental stage, whereas, *OsCRK1* expression was almost undetectable in the stages analyzed, indicating towards pseudogenization of this duplicated counter part of *OsCPK30*.

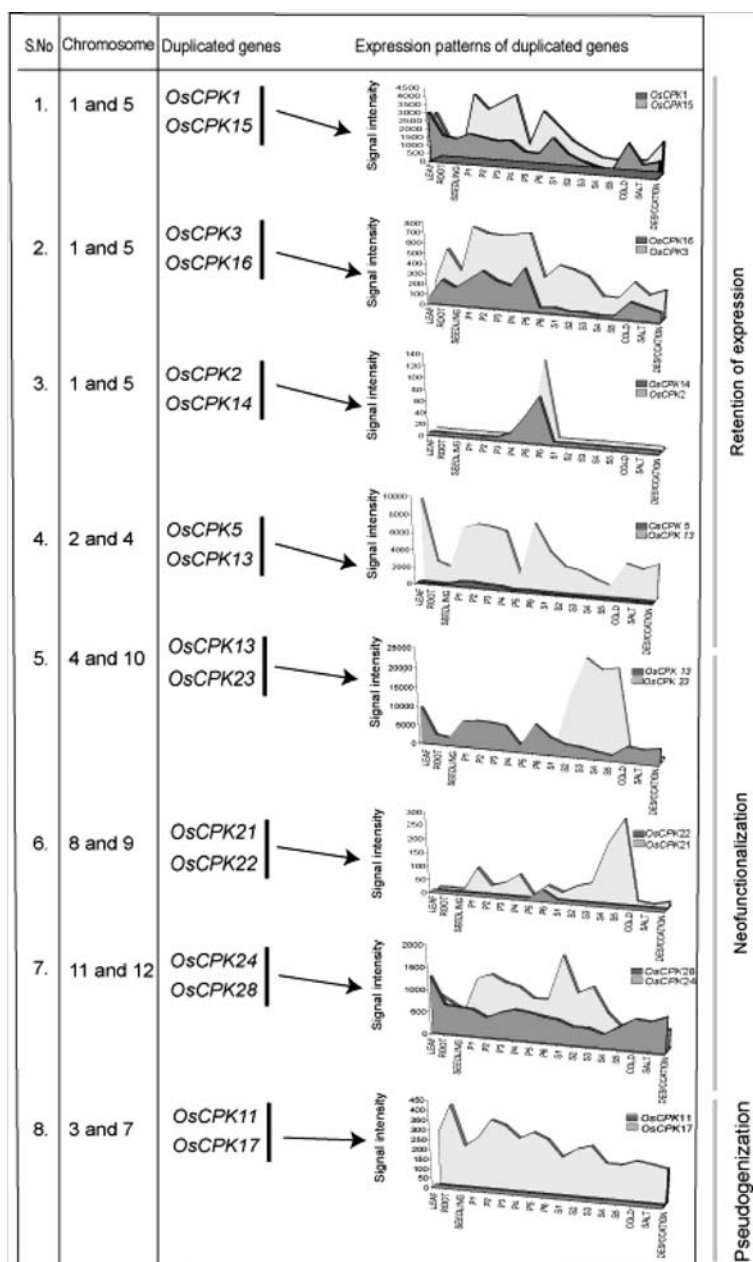
To investigate the probable explanation of this divergence in expression pattern, in-silico promoter analyses of the 1 kb upstream regions from the translation start site of the duplicated genes were carried out. With the help of PlantCARE the known regulatory elements were identified. Studies revealed that four pair of duplicated genes with similar function showed 30–46% common regulatory elements in the 1 kb upstream region analyzed. The difference in the regulatory region of these genes might have resulted

in the difference in amplitude of expression of these gene pairs. The genes with marked difference in their expression pattern had only 23–33% conservation of common regulatory elements. For example, in case of *OsCPK21*, a putative *Skn1* (GTCAT) (Takaiwa et al. 1991) and *GCN₄* binding, TGAGTCA, (Kim and Wu 1990) *cis*-regulatory element required for endosperm expression were present within the 1 kb upstream region, whereas, these motifs were absent in the 1 kb upstream region of *OsCPK22*, which does not show any expression during seed development stages. Similarly, the 1 kb upstream region of *OsCPK23* was found to contain a *cis*-acting regulatory element, *skn-1*, which could be responsible for its expression during seed developmental stages, but this element was not detected in the regulatory region of its duplicated gene pair, *OsCPK13*, hence, the loss of *OsCPK13* expression in seeds.

Discussion

During an extensive analysis of CDPK genes in rice genome, we have been able to add two new genes, namely,

Fig. 6 Comparison of the expression profiles of calcium-dependent protein kinase genes that are localized on the duplicated segment of the rice genome. The duplicated gene pairs and their respective chromosome numbers are listed. Each area graph represents compilation of the mean normalized signal intensity values from 17 stages of development/stress conditions. Gene pairs have been grouped into retention of expression, neofunctionalization and pseudogenization based on their respective profile



OsCPK30 (595 amino acid) and *OsCPK31* (584 amino acid) bearing the characteristic domains found in CDPK proteins along with one and four EF hands, respectively, to a list of 29 previously reported CDPKs genes in rice (Asano et al. 2005). *OsCPK30* was earlier reported as *OsCRK4* (Asano et al. 2005); however, we found a PROSITE predicted functional calcium binding EF-hand in the calmodulin-like domain. Therefore, this gene has been included in the CDPK gene family. CRKs, on the other hand, are described to have degenerate calcium binding sites (Harmon et al. 2001; Hrabak et al. 2003; Harper et al. 2004). Another gene, *OsCPK31*, codes for a 584 amino acid polypeptide that shows all the characteristic features of a CDPK

protein with four EF-hands and it is represented in the form of cDNA only since its location on rice genome remains to be established. Interestingly, 34 CDPK genes have been reported in *Arabidopsis* (Cheng et al. 2002; Hrabak et al. 2003), which is known to have a much smaller genome as compared to that of rice. The phylogenetic relationship between rice and *Arabidopsis* CDPKs along with CRKs, CcAMKs and PEPRKs, that are known close relatives of the CDPK class of proteins, have been previously shown by Asano et al. (2005). They had classified rice and *Arabidopsis* CDPK genes in four groups (I–IV) with two subgroups (a and b) in both group II and III. Clustering of CDPK genes in four groups, even when genes from other plant

species were added to the analysis, indicates that certain level of diversity in CDPK gene family pre-existed the divergence of monocots and dicots (Wolfe et al. 1989). Nine gene pairs from segmentally duplicated regions of rice genome (Paterson et al. 2004; The Rice Chromosomes 11 and 12 Sequencing Consortia 2005) were positioned on the phylogenetic tree in such a way that both the members of one pair were present in the same clade, reflecting an evolutionary conservation.

The expression profile of a gene is indicative of its functional relevance. Genome-wide profiling of CDPK gene family, that are important constituents of the signal transduction cascade, was performed using whole genome rice microarrays for three vegetative stages, eleven stages of panicle and seed development and three stress conditions. While stage specific expression in terms of up- or down-regulation was evident, it may be emphasized that groups of the genes that show similar expression profile during various developmental stages may or may not have similar functions because their expression could be confined to specific cell types (Liu et al. 2006). Of the eleven groups of genes classified based on expression profiles, the category 8 genes expressed in a P6 stage specific manner, which is suggestive of their expression in mature pollen. The genes are distributed in subclades IIa and IIIa of the phylogenetic tree. Among the other IIa members, *PiCDPK1*, *AtCPK17* and *34* have been shown to express in a pollen specific manner (Harper et al. 2004). Moreover, *PiCDPK1* has been implicated in pollen tube growth polarity (Yoon et al. 2006). A maize CDPK gene, which also expressed specifically during late stages of pollen development, resulted in impaired germination and growth of pollen tube upon specific inhibition of the CDPK by antisense oligonucleotide (Estruch et al. 1994). Our analysis also shows that *OsCPK2* and *14* that show homology to *PiCDPK1*, *AtCPK17* and *34* and *OsCPK25* and *26*, which have high degree of similarity to the maize CDPK are expressed during late panicle development and, thus, could be involved in similar function during pollen maturation and/or pollen tube growth. Subclade IIIa consists of three rice CDPKs along with *PiCDPK2* and *AtCPK24*, which have been shown to express in mature pollen. The *PiCDPK2* has been implicated in the regulation of pollen tube growth (Harper et al. 2004; Yoon et al. 2006), while the role of *AtCPK24* has not been ascertained as yet. Since all the subclade members show >44% identity at amino acid level (Supplementary table S8) and their expression is specifically up-regulated during later stages of panicle development, it would make them good candidates for functional validation studies targeted towards pollen development and/or pollen tube growth.

Comparison of expression profiles of the paired gene partners from segmentally duplicated regions showed either similar, differential or almost no expression in case of one

of the partner reflecting conservation, neofunctionalization or pseudogenization after the duplication event (Ohono 1970; Lynch and Conery 2000).

Seven genes namely, *OsCPK1*, *4*, *10*, *12*, *13*, *15* and *21*, were found to be regulated by abiotic stresses. *OsCPK6* and *25* have previously been reported to be up-regulated by drought and heat, respectively, whereas, *OsCPK17* has been shown to be down-regulated under cold, drought and salt stress conditions in certain stress tolerant cultivars analyzed (Wan et al. 2007). In our analysis, the amplitude of differential expression for these genes was not as significant as reported earlier, possibly due to use of different rice variety and/or experimental conditions. Most of the previously identified stress responsive CDPK genes cluster together in subclades Ia and Ib (Fig. 2). A large number of the genes in subclade Ia, viz., *ZmCPK10*, *ZmCPK11*, *NtCDPK2*, *NtCDPK3*, *OsCPK17* and *OsCPK24*, have been shown to be up-regulated in response to fungal elicitors or wounding (Murillo et al. 2001; Romeis et al. 2001; Akimoto-Tomiya et al. 2003; Szczegieliński et al. 2005; Wan et al. 2007). Amongst the rice CDPK genes which are not functionally characterized earlier, *OsCPK24* was found to be up-regulated during S1 stage of seed development, where fertilization and early embryo development occurs. It could be because the growth of pollen tubes and fungal hyphae could trigger similar response. Previously, many of the genes up-regulated during pollination/fertilization have been found to be common to those involved in defense responses, it was argued that the intrusion of pollen tube in style might be perceived to be the fungal hyphae eliciting similar responses (Lan et al. 2005). The process of pollination and wounding of style, both, have also been shown to induce the transcript level of SPP2 dioxygenase (*Solanum pollinatum pistil2*) in the ovary (Lantin et al. 1999). Hence, a thorough investigation regarding the role of these genes in biotic stress response is warranted. Amongst the subclades Ib genes with known function and/or expression profiles, *ZmCDPK1* in maize and *VrCDPK1* from mung bean were found to be up-regulated in response to low temperature and mechanical stress, respectively (Botella et al. 1996; Berberich and Kusano 1997). *OsCPK13* (*OsCDPK7*) transcripts have been shown to accumulate at enhanced levels under cold and salt stress conditions in both shoots and roots of 10-day-old seedling upon 8 and 12 h of stress treatment. Furthermore, constitutive overexpression of *OsCDPK7* conferred enhanced tolerance to cold, salt and drought stress in transgenic rice (Saijo et al. 2000). Under our assay conditions, *OsCPK13* was found to show up-regulation under all three-stress conditions, as stated above. Besides *OsCPK13*, *OsCPK5* also exhibited 6.22, 2.39 and 2.06-fold up-regulation in cold, salt and desiccation stress, respectively, but has not been included in the list of stress responsive genes because of relatively higher *P* values

(viz., 0.14, 0.34 and 0.44, respectively; see, Supplementary table S7). *OsCPK6* was up-regulated by drought (Wan et al. 2007). Since, other genes in this clade are either not functionally characterized yet or do not show differential gene expression under the stress conditions analyzed, there exist a high probability that these genes might be involved in biotic stress response or conditions other than those analyzed in present investigation.

After the perception of stress signal by the receptors, the down stream activities are triggered, which involve increased level of calcium, reactive oxygen species (ROS) and inositolphosphates. The modulation in the Ca^{2+} level is sensed by the calcium sensors (e.g., CDPKs), which trigger the phosphorylation cascade and target the major stress responsive genes or the transcription factors (Mahajan and Tuteja 2005; Vij and Tyagi 2007). It has been proposed that *OsCPK13* (*OsCDPK7*) acts at one of the branch point of cold and salt/drought responsive pathways, where it acts upstream of several late induced genes, like *rab16A*, *salt*, *wsil8* (Saijo et al. 2000). Proteomic analysis of the overexpression lines of *OsCPK7* (*OsCDPK13*) and its related proteins, calreticulin and CRTintP1 (calreticulin interacting protein 1) identified fructokinase, cytoplasmic malate dehydrogenase and alpha-tubulin to be up-regulated (Komatsu et al. 2007). These proteins may have roles in sugar sensing pathway, in repairing damage caused by cold tolerance and low-temperature stress sensor, respectively. A potato CDPK, St CDPK5, has been implicated in response to fungal elicitor that results in increased intracellular calcium (Kobayashi et al. 2007). This calcium binds to the St CDPK5 EF-hands and the activated St CDPK5 in-turn phosphorylates downstream St RBOHB (Respiratory Burst Oxidase Homolog) resulting in ROS production. Therefore, CDPKs seem to form an integral part of both biotic and abiotic stress signal transduction pathways.

The expression kinetics of the entire calcium-dependent protein kinase gene family in rice shows qualitative and quantitative changes reflecting on its functional relevance in diverse aspect of plant growth and development. While confirming the expression pattern of a few rice CDPK genes reported earlier, a much wider expression profiling exercise for the whole CDPK gene family in rice has been generated which may help select the appropriate targets for directed function analysis studies.

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